MANIPAL SCHOOL OF INFORMATION SCIENCES
MANIPAL
(A constituent unit of MAHE, Manipal)

Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Course Plan

Course Name	:	Applied Linear Algebra Lab
Course Code	:	AME 5151
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Dr. Devi Prasad M
Name of the Program Coordinator	:	Dr. Richa Sharma

Date:	8/7/2026
Signature of Program Coordinator with Date	

M. Deviprasad	
Date:	8/7/2026
Signature of Course Coordinator with Date	

Table of Contents

Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning), programare as follows

PEO No.	Education Objective
PEO 1	Acquire solid mathematical and computational skills essential for understanding, applying, and developing modern artificial intelligence and machine learning algorithms.
PEO 2	Produce industry-ready graduates with practical experience in structuring machine learning projects using state-of-the-art software.
PEO 3	Address multi-disciplinary challenges through coursework and projects by adapting to the rapidly advancing developments in artificial intelligence and machine learning approaches.

Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning), graduates will be able to:

PO 1	An ability to independently carry out research/investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	Demonstrate a degree of mastery over the area as per the specialization of the program which should be at a level higher than the requirements in an appropriate bachelor program.
PO 4	Identify appropriate and efficient algorithmic approaches for solving real-life challenges using artificial intelligence and machine learning principles and state-of-the art software prevalent in industry and academia.
PO 5	Develop teamwork and leadership skills for addressing challenges of social importance for sustainable societal development using ethical artificial intelligence and machine learning approaches.

1. Course Plan

1.1 Primary Information

Course Name	:	Applied Linear Algebra Lab [AME 5151]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Basic algebra and calculus
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No. of Contact Hours	Program Outcomes (POs)	BL
CO1	Apply Python's legacy libraries for coding vector- and matrix-related operations.	9	PO3	3
CO2	Implement models for real-life applications using linear algebra principles and analyze the results from a practical perspective.	15	PO3	3
CO3	Implement building blocks of deep learning models efficiently using linear algebra principles.	12	PO3	3

1.3 Assessment Plan

Components	Lab test	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	2 coding assignments, 20 days each	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Applying and implementing	Applying and implementing	Applying and implementing
Pattern	Answer all the questions. Maximum marks 30.	Assignment: implement and apply solutions for real-world problems using efficient matrix-vector principles.	Answer all the questions. Maximum marks 50.
Schedule	As per academic calendar.	August 20and Sepetember 20, 2026	As per academic calendar.
Topics covered	Implement and apply optimization models for real-life applications using linear algebra principles.	Understand how to perform matrix-vector operations efficiently using Python's legacy libraries; implement and apply matrix-vector principles for solving real-world problems.	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.

1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Implement Vector abstraction in Python3. Implement vector operations. Override arithmetic operators. Write unit tests.	CO1
L2	Implement Matrix abstraction in Python3. Implement matrix-vector and matrix-matrix operations. Override arithmetic operators. Write unit tests.	CO1
L3	Implement FastVector using Triton. Compare performance agsinst Vector, and PyTorch Tensor objects.	CO2
L4	Implement TiledMatrix abstraction using Triton. Compare performance	CO2
L5	Use Gensim Word2Vec model and text8 data. Test and visualize math and semantic similarities.	CO3
L6	Build a computational graph in PyTorch to implement a single layer neural network.	CO2
L7	Using the Vector and Matrix objects implement Gaussian Elimination for solving a linear system of equations.	CO3
L8	Implement Matrix Inverse method for solving a linear system of equations using Vector and Matrix classes.	CO1
L9	Implement Gram-Schmidt algorithm and QR-factorization	CO2
L10	Implement backward propagation using matrix-based approach for binary classification	CO2
L11	Implement backward propagation using matrix-based approach for multiclass classification	CO3

L12	Implement backward propagation using matrix-based approach for multiclass classification	CO3

1.5 References

1. S. Boyd and L. Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*, 1st ed. Cambridge, U.K.: Cambridge University Press, 2018. [Online]. Available: http://vmls-book.stanford.edu/vmls.pdf
2. G. Strang, *Linear Algebra and Learning from Data*, 1st ed. Cambridge, U.K.: Cambridge University Press, 2019.
3. IBM, "AI Engineering Professional Certificate," Coursera. [Online]. Available: https://www.coursera.org/professional-certificates/ai-engineer
4. F. Chollet, *Deep Learning with Python*, 2nd ed. Shelter Island, NY, USA: Manning Publications, 2021.
5. M. A. Nielsen, *Neural Networks and Deep Learning*. Determination Press, 2015. [Online]. Available: http://neuralnetworksanddeeplearning.com/
6. R. Bronson and G. B. Costa, *Matrix Methods: Applied Linear Algebra*, 3rd ed. Burlington, MA, USA: Academic Press, 2008.
7. Massachusetts Institute of Technology, "Introduction to Computational Thinking," 2024. [Online]. Available: https://computationalthinking.mit.edu/Fall24/

1.6 Other Resources (Online, Text, Multimedia, etc.)

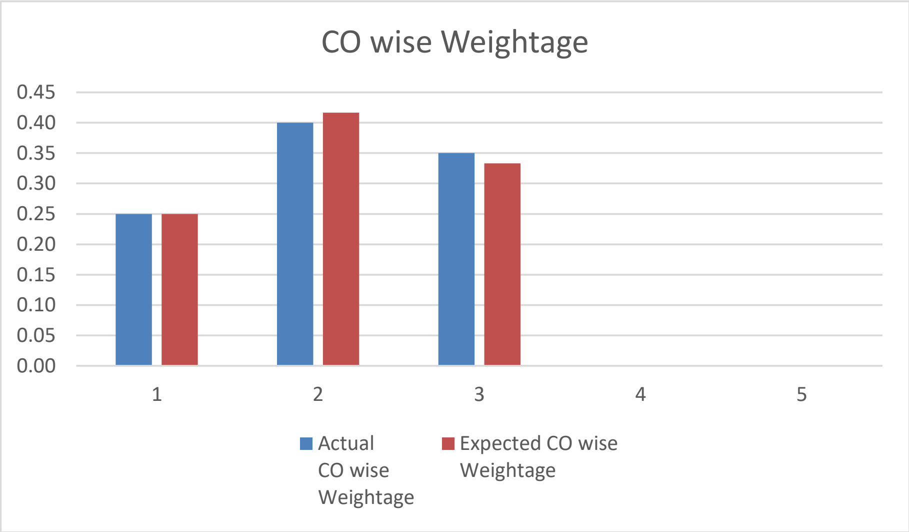
1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.

1.7 Course Timetable

Semester - I						Lab:	Embedded Systems Lab		
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday			ALA Lab						
Tuesday									
Wednesday									
Thursday									
Friday									
Saturday									

1.8 Assessment Plan

COs		Marks & Weightage				
CO No.	**CO Name**	**Lab test marks (Max. 30)**	**Assignment Marks (Max. 20)**	**ESE Marks (Max. 50)**	**Actual CO wise Weightage**	**Expected CO wise Weightage**
CO1	Apply Python's legacy libraries for coding vector- and matrix-related operations.	10	5	10	**0.25**	**0.25**
CO2	Implement models for real-life applications using linear algebra principles and analyze the results from a practical perspective.	10	10	20	**0.40**	**0.42**
CO3	Implement building blocks of deep learning models efficiently using linear algebra principles.	10	5	20	**0.35**	**0.33**
	Marks (weightage)	**0.3**	**0.2**	**0.5**	**1.00**	**1.00**



Note:

• In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes performances in lab participation, assignment work, lab work, lab test, quizzes etc.
• ESE for this course is conducted for a maximum of 50.
• ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

Weightage for CO1 = (Lab test marks for CO1 + Assignment marks for CO1 + ESE marks for CO1) /100
= (10 + 5+10)/100 = 0.25

1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

SI. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	Lab test	0.3	1	• Performance is measured using lab test attainment level. • Reference: question paper and answer scheme. • The lab test is assessed for a maximum of 30 marks.
2	Assignments	0.2	1	• Performance is measured using assignments attainment level. • Assignment is evaluated for a maximum of 20 marks.
3	ESE	0.5	1	• Performance is measured using ESE attainment level. • Reference: question paper and answer scheme. • ESE is assessed for a maximum of 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2			Y		
CO3			Y		
Average Articulation Level			*		